1 Introduction

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action taken by the agent depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

Moreover, we assume that an evidence e of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find the optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_{S} \sum_{a} \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e).$$

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg\min_{S} \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

The optimal strategy, for each evidence e, chooses the action that minimizes the expected loss conditioned on e:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg\min_{a} \mathbb{E}_{P^*} \left[\ell(a, x) \mid e \right] && (1) \\ &= \arg\min_{a} \sum_{x} \ell(a, x) P^*(x \mid e) \cdot && (2) \end{aligned}$$

This implies that the only distribution which affects the optimal strategy is $P^*(X \mid E)$. Note that the value of the expected utility depends on $P^*(E)$, but the actions that construct the optimal strategy are not affected by $P^*(E)$. Only for suboptimal strategies $P^*(E)$ plays a role and determines the importance of suboptimal actions.

Here, $P^*(X, E)$ is the joint probability distribution of the environment state and the evidence. However, how do we obtain this distribution? Is there an objective distribution inherent in the environment that we need to somehow discover?

Answering this question is not straightforward. From a subjective perspective on probability, the answer is negative. There is no such objective distribution, and P^* exists only in our minds. It serves as a tool that helps us regulate our reasoning—an intermediate measure created within the Bayesian decision-making framework to ensure we stay consistent with our assumptions and common sense principles.

Considering the subjective viewpoint, the determination of P^* relies on one's prior knowledge and assumptions. When we have no assumptions and no prior knowledge, the set of candidate probabilities for P^* simply consists of all possible probability measures. As we make assumptions or obtain some prior knowledge, those probability measures that do not align with our assumptions or knowledge are eliminated from the set. Consequently, the set of acceptable probability measures becomes more refined. We will refer to this refined set as a probabilistic model:

Definition 1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

One of the most commonly used assumptions in the statistical modeling that determines the set of acceptable probability measures is the exchangeability assumption. We say a sequence of random variables $X_1, X_2, \ldots$ is exchangeable, if the ordering of variables does not provide any useful information. In other words, for an exchangeable sequence of random variables, the joint probability is the same for any permutation of variables. This property holds true for every subset of the sequence.

For example, if each X_i represents a binary variable, the joint probability depends solely on the total count of 1s and 0s, and the specific order in which these variables appear becomes irrelevant.

When a sequence of random variables is exchangeable, de Finetti's theorem states that their joint probability has a special representation:

$$P(x_1, x_2, \ldots) = \int \prod_i \pi(x_i \mid \theta) p(\theta) d\theta.$$

This means, exchangeable observations are conditionally independent relative to some latent parameter θ.

Moreover, the theorem states that if each X_i is discrete, the distribution $\pi(x_i \mid \theta)$ corresponds to the multinomial distribution.

Definition 2 (Parametric Model)**.** Let Θ be a random variable. A *parametric model* is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{3}$$

while π is a fixed, known probability distribution. We refer to π as the *likelihood* distribution.

Now, consider our decision-making problem: Suppose the evidence E consists of a sequence of previous experiences of the environment. If we assume that our environment remains unchanged over time so that this sequence is exchangeable, to fully determine P^*, we only need to determine the prior distribution of the latent parameter $p(\theta)$.

Clearly, one approach to selecting a suitable prior distribution $p(\theta)$ is to rely on our prior knowledge about the environment. But, what if such knowledge is unavailable? Finding an answer to this important question is not simple. In Bayesian statistics, this is where non-informative priors come into play. These priors are typically chosen to be uniform or to represent a state of ignorance about the parameters. Unfortunately, these priors are not always ideal and can sometimes lead to misleading results.

There is also another consideration that could play a role in choosing an appropriate prior distribution. Suppose that in our initial decision making problem, we choose to use a different evidence instead of E, denoted as E'. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can straightforwardly answer this question. To compare any two strategies, we simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that in reality we cannot freely choose any evidence we want. We can only choose evidence that is a function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than any strategy that uses E'. Thus, at first glance, it appears that Bayesian decision theory does not favor feature selection.

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory. However, in practice we have to use feature selection because without it the original learning task requires impractical large datasets.

The key point here is that the space of strategies using E' is a subset of strategies that use E. When we are searching for the best strategy based on E, we are implicitly evaluating all simpler forms as well. For this reason, we believe if we choose an appropriate prior distribution, feature selection should be performed automatically within the Bayesian decision theory framework, when we are finding the optimal strategy. However, many commonly used non-informative priors, such as the flat prior, do not exhibit feature selection characteristics.

Throughout this discussion, we will explore a systematic approach for choosing prior distributions when prior knowledge about the environment is lacking. As we will see, these priors also exhibit some interesting feature selection characteristics.

2 Central Distribution

When we lack prior knowledge of an environment, it is still not difficult to identify some distributions that are not suitable choices for $p(\theta)$. For instance, any distribution that assigns zero probability to certain values of θ is not a good choice. If we use such a prior, evidence can never convince us that the value of θ lies within those excluded values.

We should note that, in this context, we cannot objectively assert that there is some type of underlying process in the environment for generating different values of θ that does not consider zero probability for any value of θ. Our argument is subjective. We consider some priors to be unsafe because they have some characteristics that appear to be problematic: in the obtained posterior, some values of the parameter are excluded regardless of the evidence.

We can generalize this idea and use a similar subjective approach to determine a somewhat "safe" prior distribution. Recall that our probabilistic model for the environment consists of a set of acceptable probability measures $\mathcal{P}$. Suppose that we choose a probability measure $C \in \mathcal{P}$, and determine the optimal Bayesian strategy in C. If the expected loss of this strategy in any other probability measure $P \in \mathcal{P}$ is close to the optimal strategy, we can say that using C is safe, in the sense that the optimal strategy obtained using C, is a near optimal strategy in any other acceptable probability measure in our model.

The probability measure that we choose represents our beliefs about the environment. It can be argued that this choice must depend solely on the environment. As long as the environment is the same, we should not change our choice. For example, if our loss function changes, or if the evidence that we get from the environment becomes more or less informative, none of these can change our beliefs about the environment and therefore our chosen probability measure. We will use this principle several times in this discussion and refer to it as the *principle of objectivity.*

Unfortunately, the expected loss of a strategy is a functional of the loss function. Therefore, we cannot directly use the expected loss of the optimal Bayesian strategy for evaluating different probability measures. We need a measure that quantifies the similarity between two probability measures while it ignores the specific loss function we are using.

Suppose that for an environment, we have decided on a probabilistic model $\mathcal{M}$, and without any particular prior knowledge, we have chosen a probability measure P from our model. In particular, we are interested in the distribution of a random variable X.

We ask an expert—a person familiar with the environment—to evaluate our probability measure P. Presumably, that expert has his own beliefs about the environment. Therefore, he believes, instead of P, we should use a different probability measure Q. When evaluating P, he would try to measure the loss of using the incorrect distribution $P(X)$, while he believes the correct distribution is $Q(X)$.

Let's assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left(Q_X \parallel P_X\right)$. It seems reasonable, as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}\left(Q_X \parallel P_X\right)$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose that $\mathbb{D}\left(Q_X \parallel P_X\right)$ is a divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a continuous random variable. If*

$U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have

$$\mathbb{D}\left(Q_{\Omega} \parallel U_{\Omega}\right) = \mathbb{H}\left[U_{\Omega}\right] - \mathbb{H}\left[Q_{\Omega}\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y*

$$\mathbb{D}\left(Q_{X,Y} \parallel P_{X,Y}\right) = \mathbb{D}\left(Q_X \parallel P_X\right) + \sum_x \mathbb{D}\left(Q_{Y|x} \parallel P_{Y|x}\right) Q(x)\cdot$$

If $\mathbb{D}$ satisfies these properties, then it is the Kullback-Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable Ω and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega \mid X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P_{\Omega|x}\right] = \ln P(x)$, $\mathbb{H}\left[P_{\Omega,X}\right] = 0$.

We define $Q(\Omega \mid x)$ to be an arbitrary distribution that has the same support as $P(\Omega \mid x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) &= \mathbb{H}\left[P_{\Omega,X}\right] - \mathbb{H}\left[Q_{\Omega,X}\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned} \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) &= \mathbb{H}\left[P_{\Omega|x}\right] - \mathbb{H}\left[Q_{\Omega|x}\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega\cdot \end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$
\begin{aligned}
\mathbb{D}\left(Q_X \parallel P_X\right) &= \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) - \sum_x \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) Q(x) \\
&= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x) \right) \\
&= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x) \right) \\
&= \sum_x Q(x) \ln \frac{Q(x)}{P(x)}. \qquad \square
\end{aligned}
$$

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

Thus, if Q is the probability measure that represents our beliefs, we can use $\mathbb{D}\left(Q_X \parallel P_X\right)$ to evaluate any other probability measure P, with respect to a random variable X. In other words, we can say that we are evaluating $P(X)$.

When we have two different random variables X and Y, there will be two distinct cases that should not be confused: We may want to evaluate $P(X, Y)$, or may want to evaluate $P(X)$ and $P(Y)$. Evaluating the joint distribution $P(X, Y)$ is straightforward and can be done by defining a new random variable $Z = (X, Y)$. However, the second case may present more challenges. Evaluating the marginal distributions $P(X)$ and $P(Y)$ separately is the same as evaluating $P(X, Y)$ when X and Y are independent. Here, we can use a similar argument to what we used for the additive property: We want to evaluate the loss of $P(X, Y)$, If X happens first we will face the loss of $P(X)$, then when Y happens we will face the loss of $P(Y)$ because X and Y are independent. Therefore we should use addition:

$$
\mathbb{D}\left(Q_X \parallel P_X\right) + \mathbb{D}\left(Q_Y \parallel P_Y\right).
$$

We may use the KL divergence to measure the agreement of a model about a specific distribution:

Definition 3 (ϵ-convergence)**.** In a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, let X be a discrete random variable, and $\mathcal{E}$ be a countable set of discrete random variables, all defined on $\mathcal{S}$. We refer to X as the *query variable*, and $\mathcal{E}$ as the *evidence set*.

We say $X|\mathcal{E}$ (X conditioned on $\mathcal{E}$) *ϵ-converges by* $P \in \mathcal{P}$, if for any random variable $E \in \mathcal{E}$ and any probability measure $Q \in \mathcal{P}$ we have:

$$\mathbb{E}_Q\left[\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)\right] < \epsilon,$$

where $\mathbb{E}_Q[\,\cdot\,]$ denotes expectation relative to Q and $\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)$ is the Kullback-Leibler divergence of $P(X \mid e)$ from $Q(X \mid e)$:

$$\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right) = \sum_x Q(x \mid e) \ln \frac{Q(x \mid e)}{P(x \mid e)}.$$

When $\mathcal{E}$ contains a single random variable E, We refer to E as the *evidence variable.* We also simplify our notation and instead of writing $X|\{E\}$, we use $X|E$.

ϵ-convergence is closely related to the concept of Bayesian convergence, where the posterior converges to the same distribution regardless of the prior distribution.

As an example, consider a scenario where we are trying to predict the outcome of a four-sided dice. This dice is unfair, with each face marked by a binary number: 00, 01, 10, and 11. Before making our prediction, we can roll the dice multiple times, assuming that each roll is independent.

If we let E be a vector representing the first 10 dice rolls and X be the outcome of the next roll, $X|E$ ϵ-converges by P_u for $\epsilon \geq 0.109$, where P_u is the probability measure obtained from assuming a uniform prior for the multinomial parameters. If E represents the first 20 outcomes, then we can choose any $\epsilon \geq 0.073$.

In our example, the chosen prior for the multinomial parameters affects the value of ϵ. In other words, different probability measures in our model have different convergence properties. If there is a probability measure that provides ϵ-convergence for a very small ϵ, we can hardly justify using any other probability measure.

However, that probability measure may not always exist. Moreover, we are uncertain about what value of ϵ would be sufficiently small. This becomes more problematic when we need to consider more than one distribution. For example, suppose that we are interested in the distributions of both $X|E_1$ and $X|E_2$. As we discussed earlier, when dealing with multiple distributions we should consider the sum of their KL divergences. That would make determining the appropriate value of ϵ even harder.

Definition 4 (Divergence Measure)**.** Let X be a query variable, and $\mathcal{E}$ be an evidence set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of $P \in \mathcal{P}$ for $X|\mathcal{E}$* as follows:

$$\mathbb{D}_{X|\mathcal{E}}\left[P\right] = \max_{Q \in \mathcal{P}} \max_{E \in \mathcal{E}} \mathbb{E}_Q \left[\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)\right] \tag{4}$$

$$= \max_{Q \in \mathcal{P}} \max_{E \in \mathcal{E}} \sum_{x,e} Q(X = x, E = e) \ln \frac{Q(X = x \mid E = e)}{P(X = x \mid E = e)} \cdot \tag{5}$$

Note that we usually simplify our notation and instead of writing the full assignment to a random variable, we only write the assigned value. For example, instead of $Q(X = x, E = e)$ we write: $Q(x, e)$.

Proposition. $X|\mathcal{E}$ ϵ-converges by P, if and only if:

$$\mathbb{D}_{X|\mathcal{E}}\left[P\right] < \epsilon \cdot$$

We can avoid all these hurdles by simpling choosing the probability measure that has the smallest divergence measure. Subjectively, this probability measure is the safest choice in our model and it turns out that it behaves well even when the divergence measure is not very small.

Definition 5 (Central Distribution)**.** In a model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, we define the *central distribution* for a query variable X, conditioned on the evidence set $\mathcal{E}$, to be a distribution $C_{X|\mathcal{E}} \in \mathcal{P}$ which minimizes the divergence measure for $X|\mathcal{E}$:

$$C_{X|\mathcal{E}} = \arg\min_{P \in \mathcal{P}} \mathbb{D}_{X|\mathcal{E}}\left[P\right] \cdot$$

When we need to choose one of the probability measures of our model, the central probability measure for a query set appears to be a reasonable and safe choice. However, if our query set depends on our specific experimental setup, we are violating the principle of objectivity. To avoid this, one natural solution is to let the query set include all conditional forms that are relevant.

In our dice example, we can let $X = (X_{k+1}, X_{k+2}, \ldots, X_n)$, representing all dice rolls from k to n. We then let E be a random vector of the previous dice outcomes $(X_1, X_2, \ldots, X_k)$. Here, the choice of k and n is based on our assumptions about the complexity of the problem: Only after seeing at least k outcomes, convergence is meaningful and after n outcomes it becomes trivial.

This seems to come from the fact that when we use these priors, in the resulted probability measure the estimate for the distribution of $X \mid E'$ differs from what we intuitively estimate. This estimate tends to have a much higher entropy. As a result the optimal strategy for E' is mostly ignored because for any value of the evidence it yields a poor expected utility.

It is reasonable to consider that if we find a probability measure which has a closer estimate to our intuitive estimate for $X \mid E'$ distribution, automatic feature selection will be performed within the Bayesian framework. We believe that can be done by using a probability measure that is "safe" for $X \mid E'$, in the sense we just described.

Throughout this discussion, we will further investigate this idea and try to formulate a systematic approach to determine such probability measures.

In our earlier discussion, to assess the safety of a probability measure, we realized that we need to evaluate the optimal Bayesian strategy in a generic way, independent of the utility function. We also need to determine how the divergence measures of these variables should be combined. Definition 3 already specifies the measure for a single query variable with different evidence variables. However, in this case, we have different query variables. When we have different query variables, the incorrect prediction of each variable can incur a separate loss. Therefore, we believe in this case addition should be used:

$$\sum_i \mathbb{D}$$

which equals to

$$zhoomzhooom$$

where x is the random vector and e is ...

What if we find a probability measure C that has a small divergence from any other distribution in our model? That way, every expert would agree our probability measure is a good choice.

When dealing with multiple query variables, the additivity property guides us in defining the appropriate divergence measure:

Definition 6 (Collective Divergence Measure)**.** Let X, Y be two query variables, and $\mathcal{E}_X$, $\mathcal{E}_Y$ be two evidence sets in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *collective divergence measure* of $P \in \mathcal{P}$ for $X|\mathcal{E}_X$ and $Y|\mathcal{E}_Y$ as follows:

$$\mathbb{D}_{X|\mathcal{E}_X, Y|\mathcal{E}_Y}[P] = \max_{Q \in \mathcal{P}, E_X \in \mathcal{E}_X, E_Y \in \mathcal{E}_Y} \left\{ \mathbb{E}_Q \left[\mathbb{D}\left(Q_{X|e_X} \parallel P_{X|e_X} \right) \right] + \mathbb{E}_Q \left[\mathbb{D}\left(Q_{Y|e_Y} \parallel P_{Y|e_Y} \right) \right] \right\}.$$

Let's revisit our discussion from the previous section. We saw that in Bayesian framework when we have evidence E', such that $E' = f(E)$, the optimal strategy for E always has a greater expected utility than any strategy that uses E'. We stated that we believe...

Definition 7 (Aggregator)**.** Consider two probabilistic models $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$ and $\mathcal{M}^* \equiv (\mathcal{S}^*, \mathcal{P}^*)$. Let X be a query variable and $\mathcal{E} = \{E_1, E_2, \ldots\}$ be an evidence set in $\mathcal{M}$. Also, let E^*, X^*, and I be random variables defined on $\mathcal{S}^*$ in $\mathcal{M}^*$. We refer to X^*, E^* as the *aggregated* query and evidence variables, and I as the *selector* variable.

Suppose that for a probability measure $P \in \mathcal{P}$ there exists a probability measure $P^* \in \mathcal{P}^*$, such that for any index i, and every values x, and e_i there is some x^*, and e^*, where we have

$$P(X = x, E_i = e_i) = P^*(X^* = x^*, E^* = e^* \mid I = i)\cdot \tag{6}$$

Furthermore, X^* and I are conditionally independent given E^*:

$$P^*(X^* \mid E^*, I) = P^*(X^* \mid E^*)\cdot \tag{7}$$

We call P^* an *aggregator of P with respect to $X|\mathcal{E}$.*

Proposition. By Equations 6 and 7 we have

$$P(x \mid e_i) = P^*(x^* \mid e^*)\cdot$$

Definition 8 (Aggregate Model)**.** Consider a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. Let X be a query variable, and $\mathcal{E}$ be an evidence set in $\mathcal{M}$. We say a model $\mathcal{M}^* \equiv (\mathcal{S}^*, \mathcal{P}^*)$ is the *aggregate model* of $\mathcal{M}$ with respect to $X|\mathcal{E}$, If every probability measure in $\mathcal{M}^*$ is the aggregator of a probability measure in $\mathcal{M}$ with respect to $X|\mathcal{E}$.

Definition 9 (Selective Aggregate Model)**.** Let X be a query variable, and $\mathcal{E}$ be an evidence set in $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. Let $\mathcal{M}^*$ be an aggregate model of $\mathcal{M}$ with respect to $X|\mathcal{E}$. We say $\mathcal{M}^*$ is *selective*, if for every $P \in \mathcal{P}$ and any value i of the selector variable I, there is an aggregator of P in $\mathcal{M}^*$, such that $P^*(i) = 1$.

Theorem 2. *Let X be a query variable, and $\mathcal{E}$ be an evidence set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. Suppose that $\mathcal{M}^* \equiv (\mathcal{S}^*, \mathcal{P}^*)$ is a selective aggregate model of $\mathcal{M}$ with respect to $X|\mathcal{E}$. Let X^*, E^* be the aggregated query and evidence variables in $\mathcal{M}^*$. If $P^* \in \mathcal{P}^*$ is an aggregator of the probability measure $P \in \mathcal{P}$ with respect to $X|\mathcal{E}$, then we have*

$$\mathbb{D}_{X|\mathcal{E}}\left[P\right] = \mathbb{D}_{X^*|E^*}\left[P^*\right]\cdot$$

In addition, the central distribution for $X^*|E^*$ is an aggregator of the central distribution for $X|\mathcal{E}$.

Proof. Suppose $P^* \in \mathcal{P}^*$ is an aggregator of the probability measure $P \in \mathcal{P}$ with respect to $X|\mathcal{E}$. Let

$$R^* = \arg\max_{Q^* \in \mathcal{P}^*} \sum_{x^*, e^*} Q^*(x^*, e^*) \ln \frac{Q^*(x^* \mid e^*)}{P^*(x^* \mid e^*)}.$$

Since $\mathcal{M}^*$ is an aggregate model, R^* must be an aggregator of some probability measure $R \in \mathcal{P}$. It follows that

$$\begin{aligned}
\mathbb{D}_{X^*|E^*}[P^*] &= \sum_{x^*, e^*} R^*(x^*, e^*) \ln \frac{R^*(x^* \mid e^*)}{P^*(x^* \mid e^*)} \\
&= \sum_i R^*(i) \sum_{x, e_i} R(x, e_i) \ln \frac{R(x \mid e_i)}{P(x \mid e_i)} \\
&\leq \max_i \sum_{x, e_i} R(x, e_i) \ln \frac{R(x \mid e_i)}{P(x \mid e_i)}.
\end{aligned}$$

This implies

$$\mathbb{D}_{X^*|E^*}[P^*] \leq \mathbb{D}_{X|\mathcal{E}}[P] \cdot \tag{8}$$

Similarly, let

$$(S, k) = \arg\max_{(Q,i)} \sum_{x, e_i} Q(x, e_i) \ln \frac{Q(x \mid e_i)}{P(x \mid e_i)}.$$

$\mathcal{M}^*$ is a selective aggregate model. This means that S must have an aggregator S^* in $\mathcal{M}^*$, which selects k by assigning it a probability of 1: $S^*(k) = 1$. Therefore,

$$\begin{aligned}
\mathbb{D}_{X|\mathcal{E}}[P] &= \sum_{x, e_k} S(x, e_k) \ln \frac{S(x \mid e_k)}{P(x \mid e_k)} \\
&= \sum_{x^*, e^*} S^*(x^*, e^* \mid k) \ln \frac{S^*(x^* \mid e^*)}{P^*(x^* \mid e^*)} \\
&= \sum_{x^*, e^*} S^*(x^*, e^*) \ln \frac{S^*(x^* \mid e^*)}{P^*(x^* \mid e^*)} \\
&\leq \mathbb{D}_{X^*|E^*}[P^*] \cdot
\end{aligned}$$

From this and Inequality 8, we conclude that

$$\mathbb{D}_{X|\mathcal{E}}[P] = \mathbb{D}_{X^*|E^*}[P^*] \cdot$$

To prove the second part of the theorem, let C^* be a central distribution for $X^*|E^*$. Since $\mathcal{M}^*$ is an aggregate model for $\mathcal{M}$, C^* must be an aggregator for some probability measure $C \in \mathcal{P}^*$. If we assume, by contradiction, that C is not a central distribution for $X|\mathcal{E}$ in $\mathcal{M}$, then there must be a probability measure $R \in \mathcal{P}$ such that:

$$\mathbb{D}_{X|\mathcal{E}}\left[R\right] < \mathbb{D}_{X|\mathcal{E}}\left[C\right] = \mathbb{D}_{X^*|E^*}\left[C^*\right] \cdot$$

On the other hand, since $\mathcal{M}^*$ is a selective model, R must have an aggregator $R^* \in \mathcal{P}^*$, and since C^* is a central distribution in $\mathcal{M}^*$, we have

$$\mathbb{D}_{X|\mathcal{E}}\left[R\right] = \mathbb{D}_{X^*|E^*}\left[R^*\right] \geq \mathbb{D}_{X^*|E^*}\left[C^*\right],$$

which is a contradiction. □

Definition 10 (Conditional Divergence Measure)**.** Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model. Let X be a query variable and $\mathcal{E}$ be an evidence set in $\mathcal{M}_\Theta$. We define the *divergence measure of $P \in \mathcal{P}$ given θ for $X|\mathcal{E}$* as follows:

$$\mathbb{D}_{X|\mathcal{E}}\left[P \mid \theta\right] = \max_{E\in\mathcal{E}} \mathbb{E}_\pi \left[\mathbb{D}\left(\pi_{X|e,\theta} \parallel P_{X|e}\right) \mid \theta\right] \tag{9}$$

$$= \max_{E\in\mathcal{E}} \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} \cdot \tag{10}$$

In this definition, the query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ plays a crucial role. Specifically, $\mathcal{X}$ defines the set of random variables for which we are interested in their distributions. Each X represents a random variable whose distribution is important to us, and each E corresponds to the evidence that is available for that variable.

For a parametric model (as defined in Definition 2), it seems reasonable to define a conditional form of our divergence measure, conditioned on the value of the parameter.

To do so, in Equation **??**, we replace normal expectation with conditional expectation. There will be no longer a need to maximize over different distributions, since in a parametric model the parameter fully determines the distribution of the sample space.

Even if we cannot find a distribution that has a small divergence from every distribution, still using the distribution that minimizes the maximum divergence seems a reasonable choice.

In a parametric model, the maximum of divergence happens for a prior distribution that is concentrated at a single value of the parameter θ. That means, for a parametric model which allows any high peak prior, instead of maximizing over the set of all acceptable distributions, we can simply maximize over the parameter space, using the conditional version of the divergence measure.

Lemma 1. *Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model. Let X be a query variable, and E be an evidence variable in $\mathcal{M}_\Theta$. For every $P \in \mathcal{P}$ we have*

$$\mathbb{D}_{X|E}[P] \leq \max_{\theta \in \Theta} \mathbb{D}_{X|E}[P \mid \theta]$$

Proof. Since the Kullback-Leibler divergence is positive, its expected value is also positive, and for any arbitrary $R \in \mathcal{P}$, and parameter value θ we have

$$\mathbb{E}_\pi\left[\mathbb{D}\left(\pi_{X|e,\theta} \parallel R_{X|e}\right) \mid \theta\right] = \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{R(x \mid e)} \geq 0\cdot$$

Using this, for every distribution P, we obtain

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \leq \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)}\cdot$$

Since this inequality holds for any θ, the maximum of the right-hand side w.r.t. θ is also greater than or equal to the maximum of the left-hand side:

$$\max_{\theta \in \Theta} \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \leq \max_{\theta \in \Theta} \mathbb{D}_{X|E}[P \mid \theta]\cdot \tag{11}$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x, e) = \int \pi(x, e \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a prior probability distribution. Trivially, we have

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \leq \max_{\theta \in \Theta} \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)}\cdot$$

This inequality holds for every value of θ. Therefore, we can multiply this inequality by $r(\theta)$, and integrate. Notice that the right hand side is not a function of θ and it remains unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R\left[\mathbb{D}\left(R_{X|e} \parallel P_{X|e}\right)\right] \leq \max_{\theta \in \Theta} \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)}\cdot$$

Now from Inequality 11, for *any* distribution P and R in $\mathcal{M}_\Theta$, we have

$$\mathbb{E}_R\left[\mathbb{D}\left(R_{X|e} \parallel P_{X|e}\right)\right] \leq \max_{\theta \in \Theta} \mathbb{D}_{X|E}[P \mid \theta],$$

which proves the lemma. □

Theorem 3. *Suppose $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model. Let X be a query variable, and E be an evidence variable in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:*

$$\mathbb{D}_{X|E}[P] = \max_{\theta \in \Theta} \mathbb{D}_{X|E}[P \mid \theta] \cdot \tag{12}$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let

$$\theta^* = \arg\max_{\theta \in \Theta} \mathbb{D}_{X|E}[P \mid \theta] \cdot$$

In $\mathcal{M}_\Theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(s \mid \theta^*)$ for every $s \in \mathcal{S}$. Obviously, R and $\pi(\cdot \mid \theta)$ define the same distribution for any random variable, and we have

$$\mathbb{E}_R\left[\mathbb{D}\left(R_{X|e} \parallel P_{X|e}\right)\right] = \mathbb{E}_\pi\left[\mathbb{D}\left(\pi_{X|e,\theta^*} \parallel P_{X|e}\right) \mid \theta^*\right] \cdot \tag{13}$$

On the other hand

$$\max_{Q \in \mathcal{P}} \mathbb{E}_Q\left[\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)\right] \geq \mathbb{E}_R\left[\mathbb{D}\left(R_{X|e} \parallel P_{X|e}\right)\right] \cdot$$

Notice that the left hand side is $\mathbb{D}_{X|E}[P]$, and by Equation 13

$$\mathbb{D}_{X|E}[P] \geq \max_{\theta \in \Theta} \mathbb{D}_{X|E}[P \mid \theta] \cdot$$

This inequality and Lemma 1 prove the theorem. □

Proposition. Let X be a query variable, and $\mathcal{E}$ be an evidence set in $\mathcal{M}_\Theta$. If the conditions of Theorem 3 holds, then for any $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{X|\mathcal{E}}[P] = \max_{\theta \in \Theta} \mathbb{D}_{X|\mathcal{E}}[P \mid \theta] \cdot$$

Lemma 2. *Let p, and q be two probability distributions for a random variable Θ. If for any θ, and a constant λ, we have*

$$\ln \frac{p(\theta)}{q(\theta)} = \lambda$$

then $\lambda = 0$.

Proof. Suppose, for the sake of contradiction, that $\lambda > 0$. This implies for every θ we have

$$\frac{p(\theta)}{q(\theta)} > 1\cdot$$

Since $q(\theta) > 0$, we can conclude that $p(\theta) > q(\theta)$. This inequality holds for every θ, and therefore

$$\int p(\theta)d\theta > \int q(\theta)d\theta,$$

which is a contradiction.

If we assume $\lambda < 0$, similarly we can conclude $\int p(\theta)d\theta < \int q(\theta)d\theta$, which is a contradiction again. Hence, λ cannot be positive or negative and we must have $\lambda = 0$. □

Theorem 4. *Let X be a query variable, and E be an evidence variable in a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$. Assume that each value of the parameter θ, can be represented as a pair (α, β), and for every α, β, and e we have*

$$p(\beta \mid \alpha, e) = p(\beta \mid \alpha)\cdot \tag{14}$$

If for some probability measure $P \in \mathcal{P}$, the conditional divergence of P is a real valued function of α:

$$\mathbb{D}_{X|E}\left[P \mid \theta\right] = \lambda(\alpha),$$

then P is the central distribution for $X|E$, and

$$\lambda(\alpha) = \ln \int \phi(\alpha, \beta)d\beta,$$

where

$$\phi(\alpha, \beta) = \exp\left\{\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{p(\theta \mid x, e)}{p(\alpha \mid e)}\right\}\cdot$$

Proof. By Bayes' theorem and Equation 14 we have

$$\begin{aligned}\frac{\pi(x \mid e, \theta)}{P(x \mid e)} &= \frac{p(\theta \mid x, e)}{p(\beta \mid \alpha, e)p(\alpha \mid e)} \\ &= \frac{p(\theta \mid x, e)}{p(\beta \mid \alpha)p(\alpha \mid e)}\cdot\end{aligned}$$

Now, by simple algebra we can verify that

$$\mathbb{D}_{X|E}\left[P \mid \theta\right] = \ln \frac{\phi(\alpha, \beta)}{p(\beta \mid \alpha)}$$

We let

$$\rho(\beta \mid \alpha) = \frac{\phi(\alpha, \beta)}{\int \phi(\alpha, \beta) d\beta}$$

Therefore,

$$\mathbb{D}_{X|E}\left[P \mid \theta\right] = \lambda(\alpha) + \ln \frac{\rho(\beta \mid \alpha)}{p(\beta \mid \alpha)}$$

where

$$\lambda(\alpha) = \ln \int \phi(\alpha, \beta) d\beta$$

□

Definition 11 (Generic Parametric Model)**.** We say a parametric model $\mathcal{M}_{\Theta} \equiv (\mathcal{S}, \mathcal{P})$ is *generic*, if for any arbitrary probability distribution r, there is a probability measure $R \in \mathcal{P}$, such that for every $s \in \mathcal{S}$

$$R(s) = \int \pi(s \mid \theta) r(\theta) d\theta \cdot$$

Proposition. A generic parametric model satisfies the conditions of Theorem 3.

Theorem 5 (Minimax theorem [2])**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- *$f(\cdot, y) : \mathcal{X} \to \mathbb{R}$ is concave for fixed y, and*
- *$f(x, \cdot) : \mathcal{Y} \to \mathbb{R}$ is convex for fixed x.*

Then we have that

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y) \cdot \tag{15}$$

Theorem 6. *Consider a generic parametric model $\mathcal{M}_{\Theta} \equiv (\mathcal{S}, \mathcal{P})$, and let X be a query variable, and E be an evidence variable in $\mathcal{M}_{\Theta}$. Suppose that for every probability measures $Q_1, Q_2 \in \mathcal{P}$, and any real number $0 \leq \alpha \leq 1$, there is a $Q^* \in \mathcal{P}$, such that for every x, e we have*

$$Q^*(x \mid e) = \alpha Q_1(x \mid e) + (1 - \alpha) Q_2(x \mid e) \cdot \tag{16}$$

In this model, the central distribution for $X|E$ can be obtained by:

$$C_{X|E} = \arg\max_{P \in \mathcal{P}} \int \mathbb{D}_{X|E}\left[P \mid \theta\right] p(\theta) d\theta \tag{17}$$

$$= \arg\max_{P \in \mathcal{P}} \int \sum_{x,e} p(x, e, \theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)} d\theta, \tag{18}$$

where $P(s) = \int \pi(s \mid \theta) p(\theta) d\theta$, for every $s \in \mathcal{S}$.

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X|E}) = \int \mathbb{E}_\pi \left[\mathbb{D}\left(\pi_{X|e,\theta} \parallel P_{X|e} \right) \mid \theta \right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P_{X|E}$ represents different distributions of X given different values of E, induced by a probability measure $P \in \mathcal{P}$.

Since $r(\theta)$ can be any distribution, maximizing with respect to r is similar to maximizing with respect to θ, and we have

$$C_{X|E} = \arg\min_{P\in\mathcal{P}} \max_{r} f(r, P_{X|E})\cdot$$

The domain of r is the set of all probability distributions, which is a compact convex set. Moreover, from Equation 16, it follows that the domain of $P_{X|E}$, which is a function of both X and E, is also a compact convex set. Thus, the domain of the functional $f(r, P_{X|E})$ is a compact convex set.

For a fixed $P_{X|E}$, $f(\cdot, P_{X|E})$ is a linear function, which is both concave and convex.

On the other hand, for a fixed r, if we let $R(x, e) = \int \pi(x, e \mid \theta) r(\theta) d\theta$ we have

$$f(r, \cdot) = c - \sum_{x,e} R(x, e) \ln P(x \mid e), \tag{19}$$

where c is not a function of $P_{X|E}$. If we calculate the Hessian matrix of $f(r, \cdot)$, with respect to $P_{X|E}$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x|e)}{P(x|e)^2}$. Thus, the hessian matrix of $f(r, \cdot)$ is positive semi-definite and $f(r, \cdot)$ is a convex function. Consequently, by Theorem 5

$$\min_{P\in\mathcal{P}} \max_{r} f(r, P_{X|E}) = \max_{r} \min_{P\in\mathcal{P}} f(r, P_{X|E})\cdot \tag{20}$$

From Equation 19 we can see that for an arbitrary fixed r we have

$$\min_{P\in\mathcal{P}} f(r, P_{X|E}) = \max_{P\in\mathcal{P}} \sum_{x,e} R(x, e) \ln P(x \mid e)\cdot$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\mathbb{D}\left(R_{X|e} \parallel P_{X|e} \right) \right] \geq 0\cdot$$

By simple algebra we get

$$\sum_{x,e} R(x,e) \ln P(x \mid e) \leq \sum_{x,e} R(x,e) \ln R(x \mid e)\cdot$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model. That implies $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$:

$$\min_{P\in\mathcal{P}} f(r, P_{X|E}) = \int \sum_{x,e} r(x,e,\theta) \ln \frac{\pi(x \mid e,\theta)}{R(x \mid e)} d\theta\cdot$$

By Equation 20, we conclude

$$C_{X|E} = \arg\max_{R\in\mathcal{P}} \int \mathbb{D}_{X|E}\left[R \mid \theta\right] r(\theta) d\theta\cdot \qquad \square$$

Theorem 6 requires that in our model the set of acceptable distribution for $X|E$ be a convex set. In most cases, this condition cannot be satisfied. However, if the distribution that is obtained by Equation 18 is an acceptable distribution in our model, the theorem still implies that it is a central distribution in our model. Note that the set of probability distributions is convex.

Theorem 7. *Under the assumptions of Theorem 6, the central distribution for $X|E$ satisfies:*

$$\mathbb{D}_{X|E}\left[C_{X|E} \mid \theta\right] = \lambda, \tag{21}$$

where λ is a constant.

Proof. By Theorem 6, the central distribution for $X|E$ maximizes

$$\mathcal{F}\left[p(\theta)\right] = \int \sum_{x,e} p(x,e,\theta) \ln \frac{\pi(x \mid e,\theta)}{P(x \mid e)} d\theta,$$

with respect to different choices of the prior distribution $p(\theta)$, subject to the constraint $\int p(\theta)d\theta = 1$.

Applying the Lagrange multiplier technique, the local maximum of $\mathcal{F}$ is a stationary point of the Lagrangian functional:

$$\mathcal{L}\left[p(\theta), \lambda\right] = \int \sum_{x,e} p(x,e,\theta) \ln \frac{\pi(x \mid e,\theta)}{P(x \mid e)} d\theta - \lambda \left(\int p(\theta) d\theta - 1 \right)\cdot$$

Let us denote the prior distribution corresponding to the central distribution $C_{X|E}$ by $c(\theta)$. Let $p(\theta) = c(\theta) + \epsilon\eta(\theta)$ be a variation of $c(\theta)$, where $\eta(\theta)$ is a continuously differentiable function with $\eta(0) = \eta(1) = 0$, and ϵ is a small constant.

Since $c(\theta)$ is an extremizing function for $\mathcal{F}[p(\theta)]$, the lagrangian $\mathcal{L}[\epsilon, \lambda]$ has a stationary point where $\epsilon = 0$, and we have

$$\left.\frac{\partial \mathcal{L}[\epsilon, \lambda]}{\partial \epsilon}\right|_{\epsilon=0} = 0\cdot$$

Note that

$$\begin{aligned}
\frac{dp(\theta)}{d\epsilon} &= \eta(\theta), \\
\frac{dP(x,e)}{d\epsilon} &= \int \pi(x, e \mid \theta)\eta(\theta)d\theta, \\
\frac{dP(e)}{d\epsilon} &= \int \pi(e \mid \theta)\eta(\theta)d\theta\cdot
\end{aligned}$$

Using the chain rule of calculus, we calculate the partial derivative of $\mathcal{L}[\epsilon, \lambda]$ relative to ϵ:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \epsilon} = & \int \left(\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)} \right) \eta(\theta)d\theta \\
& - \int \left(\sum_{x,e} \pi(x, e \mid \theta) \frac{\int \pi(x, e \mid \theta)\eta(\theta)d\theta}{P(x,e)} \right) p(\theta)d\theta \\
& + \int \left(\sum_{e} \pi(e \mid \theta) \frac{\int \pi(e \mid \theta)\eta(\theta)d\theta}{P(e)} \right) p(\theta)d\theta \\
& - \int \lambda\eta(\theta)d\theta
\end{aligned}$$

Since $P(x,e) = \int \pi(x, e \mid \theta)p(\theta)d\theta$ and $P(e) = \int \pi(e \mid \theta)p(\theta)d\theta$, we get

$$\frac{\partial \mathcal{L}}{\partial \epsilon} = \int \left(-\lambda + \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)} \right) \eta(\theta)d\theta\cdot$$

We let $\left.\frac{\partial \mathcal{L}}{\partial \epsilon}\right|_{\epsilon=0} = 0$. Since $\eta(\theta)$ is an arbitrary function, by applying the fundamental lemma of the calculus of variations and letting $\epsilon = 0$, for every value of θ we have

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{C_{X|E}(x \mid e)} = \lambda\cdot \qquad \square$$

Consider a scenario where we are trying to predict the outcome of a four-sided dice. This dice is unfair, with each face marked by a binary number: 00, 01, 10, and 11. Before making our prediction, we can roll the dice for several times and use the results to learn the associated probabilities. We assume that the dice rolls are independent.

Let's focus on determining the central distribution for this problem. The first step in this process is to establish our query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$.

In a prediction problem, we are interested in the distribution of the random variable we want to predict, conditioned on the observed data. For example, if we roll the dice for three times before making our prediction, we will need $P(X_4 \mid X_1, X_2, X_3)$, where X denotes the outcome of the i-th dice roll.

Thus, one natural way for defining a query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ is to let each X be the outcome of the i-th dice roll, and define each E as a vector of the previous dice outcomes $(X_1, X_2, \ldots, X_{i-1})$.

Now let us calculate the conditional divergence measure for $\mathcal{X}$ as defined in Definition 10:

$$\begin{aligned}
\mathbb{D}_{X|E}\left[P \mid \theta\right] &= \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1 \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_2 x_1 \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_3 x_1, x_2 \\
&+ \cdots \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_n x_1, \ldots, x_{n-1}.
\end{aligned}$$

After factoring out $\pi(x_1, \ldots, x_n \mid \theta)$ from every term, we use the chain rule for conditional probability:

$$\begin{aligned}
\mathbb{D}_{X|E}\left[P \mid \theta\right] &= \sum_{x_1,\ldots,x_n} \pi(x_1, \ldots, x_n \mid \theta) \ln \frac{\pi(x_1 \mid \theta)\pi(x_2 \mid x_1, \theta) \cdots \pi(x_n \mid x_1, \ldots, x_{n-1}, \theta)}{P(x_1)P(x_2 \mid x_1) \cdots P(x_n \mid x_1, \ldots, x_{n-1})} \\
&= \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1, \ldots, x_n.
\end{aligned}$$

By Equation 18, for the central distribution we have

$$
\begin{aligned}
C_{X|\mathcal{E}} &= \arg\max_P \int \left(\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1, \ldots, x_n \right) p(\theta) d\theta \\
&= \arg\max_P \int \sum_{x_1, \ldots, x_n} p(x_1, \ldots, x_n, \theta) \ln \frac{p(\theta \mid x_1, \ldots, x_n)}{p(\theta)} d\theta \cdot
\end{aligned}
$$

That implies, the central distribution for our problem is the same as the distribution we obtain when using the reference prior [1] for our main parameter.

Imagine playing a game of chance involving our four-sided dice. After rolling the dice, we input the outcome into a mysterious machine. The machine then announces whether we have won or lost the game. We have no clue how the machine works. We are not even sure if the dice outcome truly affects the machine's output. All we know is that the machine is memory-less, and the result of each game is independent.

Our goal is to learn how the machine operates through experimentation. We aim to be able to predict whether we will win the game after observing the dice outcome.

We assume that, first, we roll the dice n times, and record each outcome. Next, we input some of these dice outcomes into the machine and observe the result of each game. As we will see, this assumption is important for deriving the central distribution more easily.

Let us define a query set for this problem. We denote this set as $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$, where each X is the result of the i-th game, and the evidence E is a vector

$$
(X_1, X_2, \ldots, X_{i-1}, Y_1, Y_2, \ldots, Y_n),
$$

where each Y denotes the i-th dice outcome.

The conditional divergence for $\mathcal{X}$ can be obtained by

$$
\begin{aligned}
\mathbb{D}_{X|E}\left[P \mid \theta\right] = &\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1 y_1, \ldots, y_n \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_2 x_1, y_1, \ldots, y_n \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_3 x_1, x_2, y_1, \ldots, y_n \\
&+ \cdots
\end{aligned}
$$

After using the chain rule for conditional probability the right hand side simplifies to

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)} x_1, \ldots, x_n y_1, \ldots, y_n.$$

Consequently, an alternative way for defining our query set is to define it as a single pair $\{(\mathbf{X}, \mathbf{E})\}$, where $\mathbf{X}$ is the vector of n game's result, and $\mathbf{E}$ is the vector of n dice outcome.

Using the new definition for $\mathcal{X}$, by Equation 21 we have:

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi(x \mid e, \theta)}{C_{X|\mathcal{E}}(x \mid e)} \mathbf{xe} = \lambda$$

derive a $\theta = (\alpha, \beta)$ closed form for central dist...

$$\frac{\pi(\mathbf{x} \mid \mathbf{e}, \theta)}{C_{X|\mathcal{E}}(\mathbf{x} \mid \mathbf{e})} = \frac{\pi(\theta \mid \mathbf{x}, \mathbf{e})}{C_{X|\mathcal{E}}(\theta \mid \mathbf{e})}$$

sdfd

$$\sum_{\mathbf{x},\mathbf{e}} \pi(\mathbf{x}, \mathbf{e} \mid \alpha, \beta) \ln \frac{c_{\mathcal{X}}(\beta \mid \mathbf{x}, \mathbf{e})}{c_{\mathcal{X}}(\beta \mid \alpha)}$$

Assume that we decide to simplify our learning task. We choose to consider only the least significant digit of the dice outcome and ignore the other digit. For example, if the dice outcome is 10, we treat it as 0. This approach decreases the number of possible outcomes of the dice, making the learning process simpler. However, it also reduces our ability to predict the game's result accurately.

In order to find a central distribution for this version of our game, we should slightly modify the definition of the query set. While each X remains unchanged, we need to represent each E as a vector

$$(X_1, X_2, \ldots, X_{i-1}, Y'_1, Y'_2, \ldots, Y'_n),$$

where each Y' represents the least significant digit of the dice outcome. We denote this new query set as $\mathcal{X}'$ and the previous query set as $\mathcal{X}$.

Following similar steps as we did for $\mathcal{X}$, we can show that $\mathcal{X}'$ also has a more compact counterpart, which includes a single pair $(\mathbf{X}, \mathbf{E}')$, where $\mathbf{X}$ is the vector $(X_1, X_2, \ldots, X_n)$, and $\mathbf{E}'$ is the vector $(Y'_1, Y'_2, \ldots, Y'_n)$.

explain how the central distribution for this version is different...

When we have a lot of learning data, there is no point in using this simpler version of our learning task. However, when the available data is not enough for the more complex version, the simpler version can still produce relatively meaningful results, while the data is too small for the more complex version.

Is it possible to have a probabilistic inference framework that is able to automatically choose between these two versions of the problem based on the quality and amount of available data? It seems that the central distributing has this ability. For that, we need to use the central distribution for the query set $\mathcal{X} \cup \mathcal{X}'$. This seems reasonable, since when we want our probabilistic inference framework to choose between two versions of our problem, our query set should include random variables of both versions.

Theorem 8. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{X_1, X_2, X_3, \ldots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x \mid \theta) = \pi\left(x \mid \omega_i\right) \cdot \tag{22}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c(\theta) \propto \left(\prod_i c_{\mathcal{X}} \theta \mid \omega_i \rho(\beta \mid \alpha) \right)^{\frac{1}{|\mathcal{X}|}}, \tag{23}$$

where

$$\rho(\beta \mid \alpha) \propto \exp\left\{ \sum_x \pi(x \mid \omega_i) \ln c_{\mathcal{X}} \omega_i \mid x \right\} \cdot$$

Proof. First, we show that the set of distributions of each X is a convex set. Let $Q^*(x) = \alpha Q_1(x) + (1-\alpha) Q_2(x)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and arbitrary i, x.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x) &= \alpha \int \pi(x \mid \theta) q_1(\theta) d\theta + (1-\alpha) \int \pi(x \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta \cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha) q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 7 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_x \pi(x \mid \theta) \ln \frac{\pi(x \mid \theta)}{C_{X|\mathcal{E}}(x)} = \lambda \cdot \tag{24}$$

Moreover, from Equation 25 and Bayes' theorem it follows

$$\begin{aligned}\sum_i \sum_x \pi(x \mid \theta) \ln \frac{\pi(x \mid \theta)}{C_{X|\mathcal{E}}(x)} &= \sum_i \sum_x \pi(x \mid \omega_i) \ln \frac{\pi(x \mid \omega_i)}{C_{X|\mathcal{E}}(x)} \\ &= \sum_i \sum_x \pi(x \mid \omega_i) \ln \frac{c_{\mathcal{X}}\omega_i \mid x}{c(\omega_i)} \\ &= \sum_i \ln \frac{\rho(\beta \mid \alpha)}{c(\omega_i)}.\end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_{\mathcal{X}}\theta, \omega_i = c(\theta)$. Therefore,

$$c(\omega_i) = \frac{c(\theta)}{c_{\mathcal{X}}\theta \mid \omega_i}.$$

Now we rewrite Equation 27:

$$\begin{aligned}|\mathcal{X}| \ln c(\theta) &= \sum_i \ln\left(c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta \mid \alpha)\right) - \lambda \\ c(\theta) &= \left(e^{-\lambda} \prod_i c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta \mid \alpha)\right)^{\frac{1}{|\mathcal{X}|}}\end{aligned}$$

□

Theorem 9. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x \mid \theta) = \pi\left(x \mid \omega_i\right) \cdot \tag{25}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_{\mathcal{X}}(\beta \mid \alpha) \propto \left(\prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)\rho(\beta \mid \alpha)}{c_{\mathcal{X}}(\alpha \mid \alpha)}\right)^{\frac{1}{|\mathcal{X}|}}, \tag{26}$$

where

$$\rho(\beta \mid \alpha) \propto \exp\left\{\sum_{x,e} \pi(x, e \mid \alpha, \beta) \ln c_{\mathcal{X}}(\beta \mid x, e)\right\}.$$

Proof. First, we show that the set of distributions of each X is a convex set. Let $Q^*(x \mid e) = \alpha Q_1(x \mid e) + (1-\alpha)Q_2(x \mid e)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \le \alpha \le 1$ and arbitrary i, x, e.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned}Q^*(x \mid e) &= \alpha \int \pi(x \mid e, \beta) q_1(\beta \mid e) d\beta + (1-\alpha) \int \pi(x \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta\cdot\end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha) q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 7 the central distribution of $\mathcal{X}$ satisfies

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{C_{X|\mathcal{E}}\left(x \mid e\right)} = \lambda\cdot \tag{27}$$

Moreover, from Equation 25 and Bayes' theorem it follows

$$\begin{aligned}\lambda &= \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{C_{X|\mathcal{E}}\left(x \mid e\right)} \\ &= \sum_{x,e} \pi(x, e \mid \alpha, \beta) \ln \frac{\pi(x \mid e, \beta)}{C_{X|\mathcal{E}}(x \mid e)} \\ &= \sum_{x,e} \pi(x, e \mid \alpha, \beta) \ln \frac{c_{\mathcal{X}}(\beta \mid x, e)}{c_{\mathcal{X}}(\beta \mid e)} \\ &= \sum_{i} \ln \frac{\rho(\beta \mid \alpha)}{\beta \mid \alpha}\cdot\end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_{\mathcal{X}}\theta, \alpha, \beta = c(\theta)$ $c_{\mathcal{X}}\alpha, \alpha = c_{\mathcal{X}}\alpha$. Therefore,

$$\beta \mid \alpha = \frac{c_{\mathcal{X}}(\alpha \mid \alpha)}{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)} c_{\mathcal{X}}(\beta \mid \alpha)\cdot$$

Now we rewrite Equation 27:

$$|\mathcal{X}| \ln c_{\mathcal{X}}(\beta \mid \alpha) = -\lambda + \sum_{i} \ln \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta) \rho(\beta \mid \alpha)}{c_{\mathcal{X}}(\alpha \mid \alpha)}$$

Then:

$$
\begin{aligned}
c_{\mathcal{X}}(\beta \mid \alpha) &= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)\rho(\beta \mid \alpha)}{c_{\mathcal{X}}(\alpha \mid \alpha)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\beta \mid \alpha)\rho(\beta \mid \alpha)}{\beta \mid \alpha} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(\prod_i \frac{\rho(\beta \mid \alpha)}{\beta \mid \alpha} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(e^{-\lambda} \prod_i \frac{\rho(\alpha, \beta) c_{\mathcal{X}}(\alpha)}{c_{\mathcal{X}}(\alpha, \beta)\rho(\alpha)} \right)^{\frac{1}{|\mathcal{X}|}} \qquad \square
\end{aligned}
$$

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